

Notes: Greenstone (JPE, 2002)

The Impacts of Environmental Regulations on Industrial Activity

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1 One-sentence summary

The paper uses county-level Clean Air Act nonattainment designations and plant-level Census of Manufactures data to show that environmental regulation reduced manufacturing employment, capital, and output in regulated plants, with especially large aggregate employment losses in nonattainment counties.

2 Big picture

2.1 Question

How did the 1970 and 1977 Clean Air Act Amendments affect U.S. manufacturing activity? The paper asks whether county-level environmental regulations reduced employment, capital stock, and shipments at regulated plants.

2.2 Core empirical idea

The Clean Air Act assigns counties to pollutant-specific “nonattainment” status when ambient pollution exceeds federal standards. Plants emitting the relevant pollutant in nonattainment counties face stricter regulation than similar emitters in attainment counties.

2.3 Main finding

Nonattainment regulation reduces manufacturing activity. The preferred plant fixed-effects specification implies large losses in employment, capital stock, and shipments, particularly for carbon monoxide and ozone emitters.

2.4 Why it matters

The paper is one of the first large-scale plant-level estimates of the economic costs of environmental regulation. It quantifies the production-side burden of air quality rules rather than only focusing on pollution reductions or health benefits.

3 Incremental contribution of the paper

3.1 Plant-level evidence on regulation

The paper moves beyond state- or industry-level comparisons by using confidential plant-level Census of Manufactures data. This allows the author to compare regulated and unregulated plants while controlling for plant, county, industry, and period factors.

3.2 Pollutant-specific treatment design

The paper exploits the fact that nonattainment status differs by pollutant. A plant is treated only if it emits the pollutant for which its county is in nonattainment. This makes the design sharper than a broad county-level regulation indicator.

3.3 Clean Air Act as quasi-experiment

The 1970 and 1977 Clean Air Act Amendments generated national standards but local enforcement variation through county nonattainment designations. This creates regulatory variation across counties, pollutants,

and periods.

3.4 Multiple outcomes

The paper studies employment, capital stock, and shipments. This makes it possible to interpret whether regulation affects labor demand alone or broader plant scale.

3.5 Magnitude of aggregate costs

The paper translates regression coefficients into aggregate job, capital, and shipment losses. This connects micro estimates to policy-relevant economic costs.

4 Data and measurement

4.1 Census of Manufactures plant data

The main production data come from the 1967, 1972, 1977, 1982, and 1987 Census of Manufactures. The panel includes plant-level employment, capital stock, and shipments. The analysis focuses on manufacturing plants whose emissions can be linked to regulated pollutants.

4.2 Nonattainment status

A county is designated nonattainment for a pollutant if ambient air quality exceeds the National Ambient Air Quality Standard. The paper tracks four pollutants: carbon monoxide, ozone, sulfur dioxide, and total suspended particulates.

4.3 Pollutant emitters

Industries are mapped to emissions of pollutants. Treatment requires both: (i) the plant belongs to an industry emitting the relevant pollutant and (ii) the plant's county is in nonattainment for that pollutant.

4.4 Outcome variables

The main outcomes are five-year percentage changes in employment, capital stock, and shipments. The paper emphasizes plant growth rates rather than levels, allowing plant fixed effects to absorb permanent differences in size and productivity.

5 Empirical specification

5.1 Baseline treatment effect

The core treatment variable is pollutant-specific nonattainment status for plants emitting that pollutant. A simplified version of the growth regression is:

$$\Delta y_{i,c,j,t} = \beta_p \cdot Nonattain_{c,p,t-1} \cdot Emitter_{j,p} + \Gamma X_{i,c,j,t} + FE + \varepsilon_{i,c,j,t}.$$

Here i indexes plants, c counties, j industries, p pollutants, and t periods.

5.2 Fixed effects structure

The preferred models include industry-by-period effects, county-by-period effects, and plant fixed effects. These controls absorb national industry shocks, local county shocks, and persistent plant heterogeneity.

5.3 Interpretation of coefficients

The coefficient β_p is the estimated effect of being an emitter of pollutant p in a county that was nonattainment for pollutant p at the start of the period. A negative coefficient means regulated plants grew more slowly than comparable unregulated plants.

5.4 Magnitude calculations

The paper multiplies estimated treatment effects by baseline activity in affected plants to calculate aggregate losses in employment, capital stock, and shipments.

6 Identification and empirical design

6.1 Source of variation

Identification comes from changes in pollutant-specific county nonattainment status over time and differences in industry pollutant emissions. The design compares emitters of a pollutant in nonattainment counties to similar emitters in attainment counties, while controlling for county and plant fixed effects.

6.2 Key identifying assumption

Conditional on the fixed effects and controls, changes in pollutant-specific nonattainment status are not correlated with unobserved shocks to growth in emitting plants. In other words, absent regulation, emitting plants in nonattainment and attainment counties would have followed comparable growth paths.

6.3 Why pollutant-specific comparisons help

A county may be nonattainment for one pollutant but not another. A plant emitting a regulated pollutant is treated, whereas plants emitting other pollutants in the same county may not be treated. This structure helps separate regulation from general local economic shocks.

6.4 Why plant fixed effects matter

Plant fixed effects absorb permanent differences across plants, such as size, technology, managerial quality, or location-specific productivity. The preferred estimates therefore use within-plant changes over time.

6.5 Robustness logic

The paper checks dynamic specifications, samples restricted to continuing plants, and alternative emission-rule definitions. The robustness tests ask whether the baseline effects are driven by plant exit, lagged trends, or the definition of emitters.

6.6 Limitations

Nonattainment is not randomly assigned. Pollution status is related to industrial concentration and urbanization. The strength of the design comes from rich fixed effects and pollutant-specific variation, but unobserved time-varying plant shocks remain a possible concern.

7 Tables and figures

7.1 Table 1: Incidence and Changes in Nonattainment Status; Figures 1–4: Pollutant-specific nonattainment maps

Read: Table 1 counts counties by pollutant and period, separating continuing nonattainment counties from those newly entering or leaving nonattainment. Figures 1–4 map the number of periods in which each county is in nonattainment for carbon monoxide, ozone, sulfur dioxide, and TSPs.

Economic message: The table and maps show that regulatory exposure varies substantially across pollutants, counties, and time. This variation is the foundation of the empirical strategy.

Interpretation: Ozone and TSP nonattainment are much more widespread than carbon monoxide or sulfur dioxide. The maps also reveal that nonattainment is not confined to one region; it appears in industrial corridors, urban areas, and parts of California and the Northeast.

	Nonattainment Period t (1)	Attainment Period $t-1$ and Nonattainment Period t (2)	Nonattainment Period $t-1$ and Attainment Period t (3)
A. Carbon Monoxide (CO)			
1967–72	0	0	0
1972–77	81	81	0
1977–82	144	90	27
1982–87	137	15	22
B. Ozone (O ₃)			
1967–72	0	0	0
1972–77	32	32	0
1977–82	626	595	1
1982–87	560	104	170
C. Sulfur Dioxide (SO ₂)			
1967–72	0	0	0
1972–77	34	34	0
1977–82	87	75	22
1982–87	60	7	34
D. Total Suspended Particulates (TSPs)			
1967–72	0	0	0
1972–77	296	296	0
1977–82	235	108	169
1982–87	176	24	83

NOTE.—There are 3,070 counties in the Census of Manufactures data files. See the Data Appendix for a description of how the pollutant-specific nonattainment designations are assigned.

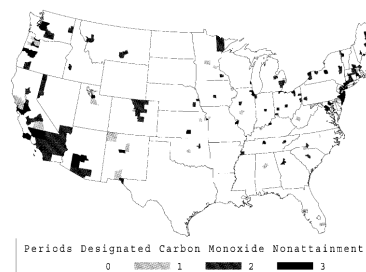


FIG. 1.—Incidence of nonattainment for carbon monoxide by county (1972–77, 1977–82, and 1982–87). Source: EPA Air Quality System Database, Code of Federal Regulations (various issues).

- (a) Table 1: Incidence and changes in nonattainment status. (b) Figure 1: Carbon monoxide nonattainment by county.

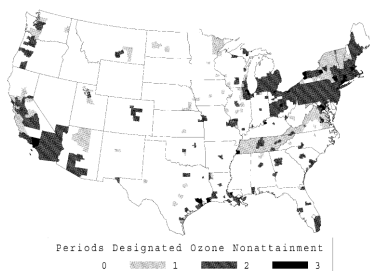


FIG. 2.—Incidence of nonattainment for ozone by county (1972–77, 1977–82, and 1982–87). Source: EPA Air Quality Subsystem Database, *Code of Federal Regulations* (various issues).

(a) Figure 2: Ozone nonattainment by county.

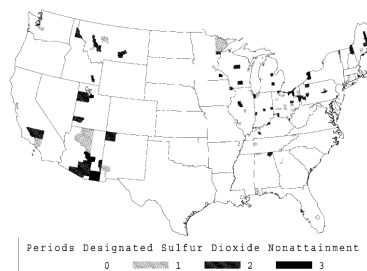


FIG. 3.—Incidence of nonattainment for sulfur dioxide by county (1972–77, 1977–82, and 1982–87). Source: EPA Air Quality Subsystem Database, *Code of Federal Regulations* (various issues).

(b) Figure 3: Sulfur dioxide nonattainment by county.

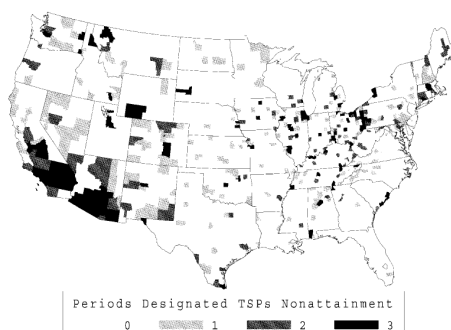


FIG. 4.—Incidence of nonattainment for total suspended particulates by county (1972–77, 1977–82, and 1982–87). Source: EPA Air Quality Subsystem Database, *Code of Federal Regulations* (various issues).

Figure 4: TSP nonattainment by county. Single unpaired figure uses width 0.6\linewidth by design.

7.2 Table 2: Manufacturing Employment by Pollutant Emitted and Pollutant-specific Attainment Status; Table 3: Means of County and Plant Characteristics

Read: Table 2 reports the scale of manufacturing employment in emitting plants by pollutant and attainment status. Table 3 compares county and plant characteristics for carbon monoxide attainment and nonattainment groups.

Economic message: Regulation affects a large share of manufacturing activity, especially ozone and TSP emitters. Table 3 shows why controls are necessary: nonattainment counties differ from attainment counties in population, income, urbanization, and plant composition.

Interpretation: The balance table is not meant to prove random assignment. It shows the observable differences that must be addressed with fixed effects and controls. Nonattainment counties are more urban and industrial, so raw comparisons would be misleading.

TABLE 2
MANUFACTURING EMPLOYMENT, BY POLLUTANT EMITTED AND POLLUTANT-SPECIFIC
ATTAINMENT STATUS

	1967-72 (1)	1972-77 (2)	1977-82 (3)	1982-87 (4)
CO-emitting plants	1,111,534	1,040,563	951,515	744,061
CO attainment	1,111,534	839,456	648,526	517,767
CO nonattainment	...	201,108	302,989	226,294
O ₃ -emitting plants	5,453,418	5,581,151	5,542,548	5,412,151
O ₃ attainment	5,453,418	5,108,078	1,294,500	1,492,627
O ₃ nonattainment	...	473,073	4,248,048	3,919,524
SO ₂ -emitting plants	1,783,243	1,717,904	1,598,742	1,358,083
SO ₂ attainment	1,783,243	1,468,781	1,233,592	1,170,479
SO ₂ nonattainment	...	249,123	365,150	187,604
TSPs-emitting plants	2,101,561	2,071,924	1,899,173	1,697,843
TSPs attainment	2,101,561	1,303,442	1,114,749	1,160,430
TSPs nonattainment	...	768,482	784,424	537,413
Total manufacturing sector	17,438,187	17,350,726	17,521,355	17,100,413

NOTE.—See the note to table 1. Employment is the mean of total employment in the first and last years of each five-year period covered by the 1967-87 Censuses of Manufactures.

TABLE 3
MEANS OF COUNTY AND PLANT CHARACTERISTICS BY 1972-77 CO NONATTAINMENT
STATUS

	CO Attainment, 1972-77 (1a)	CO Attainment, 1972-77, and CO Nonattainment, 1977-82 or 1982-87 (1b)	CO Nonattain- ment, 1972-77 (2)
A. County Characteristics in 1970			
Number of counties	2,989	100	81
Population	47,157	395,376	620,654
Population density	1,826	6,354	4,868
% urban	.65	.90	.94
% ≥12 years of education	.50	.55	.57
% ≥16 years of education	.10	.11	.13
% employment in manufacturing	.262	.266	.242
Unemployment rate	.044	.045	.046
Poverty rate	.119	.082	.081
Income per capita (1982-84 dollars)	7,456	8,712	9,414
Per capita government revenues	248	296	403
B. CO-Emitting Plant Characteristics in 1972			
Number of CO-emitting plants	1.0	6.8	14.2
Average employment	269	362	175
% operating at least 10 years	55.2	59.3	51.3
% part of multiunit firm	34.6	40.7	40.1

NOTE.—See the note to table 1. All entries are averages across counties in the relevant category. The data on county characteristics are derived from the 1970 Census. The 1972 Census of Manufactures is used to determine the means of CO-emitting plant characteristics. The entries in col. 1a are calculated from the 2,989 counties that are designated CO attainment in the 1972-77 period, and the sample in col. 2 comprises the 81 counties that are CO nonattainment in the same period. Col. 1b is the subset of the col. 1a counties that are CO attainment in 1972-77 and CO nonattainment in at least one of the 1977-82 and 1982-87 periods.

(a) Table 2: Employment levels by pollutant and attainment status. (b) Table 3: County and plant characteristics by CO status.

7.3 Table 4: Estimated Regression Models for Employment with One Regulation Effect per Regression; Table 5: Estimated Regression Models for Employment

Read: Table 4 estimates one pollutant effect at a time. Table 5 estimates a joint model with all pollutant-specific nonattainment effects and richer fixed effects.

Economic message: The employment effects are generally negative, especially for carbon monoxide and ozone. The preferred specification in Table 5 uses plant fixed effects and indicates economically meaningful employment reductions.

Interpretation: Table 4 is useful for seeing pollutant-specific patterns separately. Table 5 is the main employment table because it controls for multiple pollutant designations and includes the most demanding fixed effects.

TABLE 4
ESTIMATED REGRESSION MODELS FOR THE PERCENTAGE CHANGE IN EMPLOYMENT WITH ONE REGULATION EFFECT PER REGRESSION

	CARBON MONOXIDE		OZONE		SULFUR DIOXIDE		TOTAL SUSPENDED PARTICULATES	
	CO Emitters (N=14,556) (1)	All Plants (N=1,620,942) (2)	O ₃ Emitters (N=543,121) (1)	All Plants (N=1,620,942) (2)	SO ₂ Emitters (N=95,854) (1)	All Plants (N=1,620,942) (2)	TSPs Emitters (N=257,155) (1)	All Plants (N=1,620,942) (2)
O ₃ regulation effect	-.041 (.040)	-.074 (.031)						
S ₂ regulation effect			.068 (.011)	.025 (.009)				
O ₃ regulation effect					-.049 (.030)	-.040 (.027)		
TSPs regulation effect							-.021 (.017)	-.016 (.014)
R ²	.127	.100	.112	.100	.095	.100	.121	.109

NOTE.—The entries are taken from regressions in which the dependent variable is the change in plant employment between t and $t+5$, divided by the mean of the t and $t+5$ levels. The equations are weighted by the denominator of the dependent variable. All specifications include county fixed effects and industry by period indicators. Heteroskedastic-consistent standard errors are reported in parentheses.

(a) Table 4: Employment effects with one regulation effect per regression.

TABLE 5
ESTIMATED REGRESSION MODELS FOR THE PERCENTAGE CHANGE IN EMPLOYMENT

	(1)	(2)	(3)	(4)
CO regulation effect (β_1)	-.084 (.032)	-.075 (.031)	-.086 (.030)	-.163 (.045)
O ₃ regulation effect (β_2)	.001 (.011)	.022 (.010)	-.011 (.010)	-.049 (.015)
SO ₂ regulation effect (β_3)	-.004 (.029)	-.016 (.028)	.003 (.029)	.001 (.036)
TSPs regulation effect (β_4)	-.024 (.014)	-.010 (.013)	-.020 (.013)	-.024 (.024)
R ²	.109	.119	.144	.504
Industry by period fixed effects	yes	yes	yes	yes
Nonattainment by period fixed effects	yes	yes	no	no
County fixed effects	no	yes	no	no
County by period fixed effects	no	no	yes	yes
Plant fixed effects	no	no	no	yes

NOTE.—See the note to table 4. In all specifications, the sample includes the 1,620,942 plant observations with nonmissing and nonnegative employment levels. The mean five-year growth rate of employment in the sample is -1.4 percent.

(b) Table 5: Employment effects with alternative fixed effects.

7.4 Table 6: Capital Stock and Shipments; Table 7: Industry-specific Regulation Effects

Read: Table 6 shows regulation effects on capital stock and shipments. Table 7 asks whether employment effects vary across industries.

Economic message: The Clean Air Act affected more than employment. Regulated plants also reduced capital growth and shipments, implying a contraction in plant scale.

Interpretation: The Table 6 preferred estimates indicate negative effects on capital and shipments for several pollutants. Table 7 shows that the effects are not driven by a single industry; rather, they appear across several emitting industries.

TABLE 6
ESTIMATED REGRESSION MODELS FOR THE PERCENTAGE CHANGE IN CAPITAL STOCK AND SHIPMENTS

	(1)	(2)	(3)	(4)
A. Capital Stock (N=1,607,332)				
CO regulation effect (β_1)	-.047 (.043)	-.047 (.042)	-.097 (.043)	-.092 (.062)
O ₃ regulation effect (β_2)	-.009 (.022)	.016 (.021)	-.001 (.021)	-.041 (.029)
SO ₂ regulation effect (β_3)	-.024 (.047)	-.048 (.049)	-.057 (.055)	-.063 (.048)
TSPs regulation effect (β_4)	.026 (.027)	.042 (.025)	.010 (.024)	-.043 (.039)
R ²	.074	.109	.155	.462
B. Shipments (N=1,737,753)				
CO regulation effect (β_1)	-.058 (.029)	-.036 (.029)	-.072 (.029)	-.146 (.046)
O ₃ regulation effect (β_2)	.022 (.018)	.048 (.018)	.019 (.016)	-.032 (.024)
SO ₂ regulation effect (β_3)	-.007 (.033)	-.026 (.030)	-.027 (.030)	-.010 (.039)
TSPs regulation effect (β_4)	-.014 (.019)	-.002 (.018)	-.010 (.018)	-.032 (.034)
R ²	.127	.142	.185	.516
Industry by period fixed effects	yes	yes	yes	yes
Nonattainment by period fixed effects	yes	yes	no	no
County fixed effects	no	yes	no	no

(a) Table 6: Effects on capital stock and shipments.

TABLE 7
DO THE EMPLOYMENT REGULATION EFFECTS VARY BY INDUSTRY?

Industry Name (SIC Code)	CO Regulation Effects (1)	O ₃ Regulation Effects (2)	SO ₂ Regulation Effects (3)	TSPs Regulation Effects (4)
Lumber and wood (24)				-.006 (.034)
Pulp and paper (2611-31)	-.080 (.077)	-.110 (.056)	-.105 (.074)	.006 (.064)
Iron and steel (3312-13, 3321-25)	-.177 (.061)	-.104 (.068)	.038 (.059)	-.012 (.050)
Printing (2711-89)		-.072 (.027)		
Organic chemicals (2961-69)		.071 (.151)		
Rubber and plastic (30)		-.093 (.046)		
Fabricated metals (34)		-.013 (.026)		
Motor vehicles (371)		-.026 (.057)		
Inorganic chemicals (2812-19)			-.089 (.113)	
Petroleum refining (2911)	-.133 (.092)	.172 (.101)	-.180 (.109)	
Stone, clay, and glass (32)		-.072 (.039)	.039 (.062)	-.063 (.039)
Nonferrous metals (333-34)	-.169 (.163)		-.063 (.147)	
χ^2 statistic of equality	1.03 (.79)	11.67 (.17)	5.82 (.32)	1.57 (.67)

NOTE.—See the note to table 5. All the entries are taken from a single regression in which the dependent variable is the change in plant employment between t and $t+5$, divided by the mean of the t and $t+5$ levels. The specification

(b) Table 7: Industry-specific regulation effects.

7.5 Table 8: Robustness; Table 9: Magnitudes

Read: Table 8 reports robustness checks using dynamic models, samples restricted to continuing plants, and alternative emission rules. Table 9 converts coefficient estimates into aggregate economic magnitudes.

Economic message: Robustness checks broadly support the negative regulation effects. Table 9 shows the policy relevance: the estimates imply large employment, capital, and shipment losses over the 1972–87 period.

Interpretation: Table 9 is the bridge from micro regression estimates to aggregate cost accounting. It reports an implied employment loss of roughly 591,900 jobs for all manufacturing, a capital stock loss of about 36.6 billion, and shipment losses of about 75.3 billion in 1987 dollars.

TABLE 8
PROBING THE ROBUSTNESS OF THE REGULATION EFFECTS

	Base Specification (0)	Dynamic Model (1)	Limit Sample to "Stayers" (2)	4.5% Emission Rule (3)
A. Total Employment				
CO regulation effect (β_4)	-.086 (.030)	-.094 (.028)	-.059 (.023)	-.097 (.028)
O ₃ regulation effect (β_5)	-.011 (.010)	-.007 (.010)	-.019 (.008)	-.016 (.010)
SO ₂ regulation effect (β_6)	.003 (.029)	.005 (.027)	.010 (.021)	.006 (.028)
TSPs regulation effect (β_7)	-.020 (.013)	-.013 (.014)	-.022 (.011)	-.013 (.013)
B. Capital Stock				
CO regulation effect (β_4)	-.097 (.043)	-.134 (.041)	-.110 (.033)	-.115 (.040)
O ₃ regulation effect (β_5)	-.001 (.021)	-.007 (.021)	-.021 (.016)	-.009 (.020)
SO ₂ regulation effect (β_6)	-.057 (.055)	-.085 (.045)	-.032 (.036)	-.006 (.052)
TSPs regulation effect (β_7)	.010 (.024)	.002 (.024)	-.038 (.021)	.010 (.033)
C. Shipments				
CO regulation effect (β_4)	-.072 (.029)	-.092 (.027)	-.048 (.024)	-.075 (.027)
O ₃ regulation effect (β_5)	.019 (.016)	-.019 (.016)	.000 (.015)	.016 (.016)
SO ₂ regulation effect (β_6)	-.027 (.030)	-.054 (.025)	-.023 (.025)	-.020 (.030)
TSPs regulation effect (β_7)	-.010 (.018)	-.054 (.016)	-.037 (.015)	.008 (.020)

NOTE.—See the notes to tables 5 and 6. The entries are the estimated regulation effects and heteroskedastic standard errors (in parentheses) from separate regressions for the three dependent variables. The dependent variable is identified in the panel heading. Each column represents a different specification or sample. All specifications include county by period effects and industries by period effects. The entries in col. 0 are taken from col. 8 of table 5 (employment) and col. 0 of table 6 (capital stock and shipments).

(a) Table 8: Robustness checks.

TABLE 9
TWO MEASURES OF THE MAGNITUDE OF THE REGULATION EFFECTS

	ESTIMATED REGULATION-INDUCED CHANGE, 1972-77 TO 1982-87	CHANGE 1972-77 TO 1982-87	MEAN OF 1972-77 AND 1982-87 LEVELS	RATIO OF COL. 1 TO COL. 3	RATIO OF COL. 1 TO COL. 4
	Mean (1)	95% Confidence Interval (2)	(3)	(4)	(5)
A. Total Employment					
CO emitters	-119,100	[-54,600, -183,500]	-296,502	892,312	402
O ₃ emitters	-423,400	[-169,400, -677,400]	-169,000	5,496,051	2,505
SO ₂ emitters	800	[57,400, -55,800]	-359,821	1,537,994	-.002
TSPs emitters	-50,200	[48,200, -148,500]	-374,081	1,884,883	.134
All manufacturers	-591,900	[-118,400, -1,065,200]	-250,183	17,215,016	2,366
B. Capital Stock (Millions of Dollars)					
CO emitters	-7,500	[2,400, -17,500]	65,977	110,639	-.114
O ₃ emitters	-18,600	[7,200, -44,300]	175,235	258,645	-.106
SO ₂ emitters	-4,800	[2,400, -11,900]	85,092	144,078	-.056
TSPs emitters	-5,700	[4,500, -15,900]	56,635	108,261	-.101
All manufacturers	-36,600	[16,400, -89,600]	409,687	565,888	-.089
C. Shipments (Millions of 1987 Dollars)					
CO emitters	-25,700	[-9,800, -41,500]	-25,601	235,616	1,003
O ₃ emitters	-40,500	[19,000, -100,000]	2,281	773,443	-17,751
SO ₂ emitters	-1,500	[10,000, -13,000]	-29,806	310,140	.050
TSPs emitters	-7,600	[8,200, -23,500]	-24,581	211,875	.310
All manufacturers	-75,300	[27,400, -178,000]	227,673	2,051,492	-.331

NOTE.—The entries in col. 1 are calculated by multiplying the parameter estimates from col. 4 of tables 5 (employment) and 6 (capital stock and shipments) and the level of the outcomes in emitters in nonattainment counties; table 2 presents the employment levels. For instance, the effect of CO regulation on employment in CO-emitting industries is calculated by multiplying the estimated effect of CO nonattainment (–.163) by the sum of the levels of employment in CO-emitting plants located in CO nonattainment counties for 1972–77 (201,108), 1977–82 (302,989), and 1982–87 (226,294), which yields an estimated change of –119,100 jobs. Col. 2 presents the 95 percent confidence interval of

(b) Table 9: Two measures of the magnitude of regulation effects.

7.6 Appendix Table A1: Selected National Ambient Air Quality Standards; Appendix Table A2: Annual Industrial Sector Pollutant Releases by Industry

Read: Table A1 lists the federal standards used to determine nonattainment. Table A2 maps industries to pollutant releases and emitter status.

Economic message: These appendix tables define the regulatory and emissions measurement inputs used in the treatment assignment.

Interpretation: Table A1 explains the threshold rules behind nonattainment status. Table A2 shows that pollutant-emitting status is industry-specific, which allows the paper to compare emitters and nonemitters within local areas.

TABLE A1
SELECTED NATIONAL AMBIENT AIR QUALITY STANDARDS

	Maximum Allowable Concentration (Primary Standard)
Carbon monoxide:	
Maximum 8-hour concentration	9 parts per million
Maximum 1-hour concentration	35 parts per million
Nitrogen dioxide:	
Annual arithmetic mean	.053 parts per million
Ozone:	
Maximum 1-hour concentration	.12 parts per million (after 1979)
	.08 parts per million (through 1979)
Sulfur dioxide:	
Annual arithmetic mean	.03 parts per million
Maximum 24-hour concentration	.14 parts per million
Total suspended particulates:	
Annual geometric mean	75 micrograms per cubic meter
Maximum 24-hour concentration	260 micrograms per cubic meter

NOTE.—A county is in violation of one of the hourly based standards (i.e., one-hour, eight-hour, or 24-hour) if it exceeds the standard more than once in a year. In 1987 the EPA switched its focus from the regulation of all particulates (i.e., TSPs) to small particulates (i.e., PM₁₀s). In 1997 the ozone standard was revised, and the particulates standard

TABLE A2
ANNUAL INDUSTRIAL SECTOR POLLUTANT RELEASES BY INDUSTRY

INDUSTRY (SIC Code)	CARBON MONOXIDE		NITROGEN DIOXIDE		VOLATILE ORGANIC COMPOUNDS		SULFUR DIOXIDE		TOTAL NON-FERROUS PARTICULATES		EMISSION STATUS
	Emissions (1)	Share (2)	Emissions (1)	Share (2)	Emissions (1)	Share (2)	Emissions (1)	Share (2)	Emissions (1)	Share (2)	
Metal mining (10)	5,591	.2%	28,583	1.6%	1,283	.1%	84,222	3.5%	140,032	13.4%	*
Nonmetal mining (14)	4,522	.1%	20,804	1.0%	1,756	.2%	24,129	1.0%	107,048	10.2%	*
Lumber and wood products (24)	123,756	3.5%	42,038	2.4%	41,423	5.0%	9,149	.4%	63,761	7.0%	TSPs
Food and kindred products (20)	2,069	.1%	2,081	.2%	59,426	4.4%	1,606	.1%	3,178	.3%	Clean
Textile mill (22)	624,291	17.5%	394,448	21.7%	96,875	7.1%	941,002	14.0%	113,371	12.5%	CO ₂ /VOCs/TSPs
Printing (27)	8,463	.2%	4,913	.3%	101,537	7.5%	1,728	.1%	1,031	.1%	O ₃
Chemical and allied products (28)	186,147	4.7%	108,575	6.0%	32,091	3.0%	182,169	7.5%	30,682	4.3%	SO ₂
Plastics, rubber, and miscellaneous plastic products (30)	146,947	4.1%	256,829	13.9%	201,888	14.8%	132,459	5.4%	44,960	4.9%	O ₃
Petroleum refining (33)	419,311	11.8%	380,641	21.0%	309,058	22.7%	648,153	26.6%	36,077	4.1%	CO ₂ /VOCs/NO _x
Nonferrous metal (3325-34)	2,090	.1%	11,914	.7%	140,741	10.3%	29,361	1.2%	3,355	.0%	O ₃
Stone, clay, glass, and concrete (32)	50,043	1.0%	330,482	18.6%	30,282	2.2%	339,216	13.9%	171,833	18.0%	CO ₂ /VOCs/TSPs
Iron and steel (3325-33, 3325-25)	1,518,642	42.6%	138,085	7.7%	82,292	6.0%	238,268	9.8%	83,017	9.1%	CO ₂ /VOCs/NO _x
Nonferrous metal (3325-34)	448,758	12.6%	55,658	3.1%	27,273	2.0%	373,067	15.3%	22,000	2.3%	CO ₂ /VOCs
Fabricated metal (34)	3,831	.1%	16,424	.9%	102,186	7.5%	4,019	.2%	3,136	.3%	O ₃
Electronics (36)	807	.0%	1,129	.1%	4,854	.4%	453	.0%	293	.0%	Clean
Motor vehicles, bodies, and parts (37)	35,303	1.0%	23,725	1.3%	101,275	7.4%	25,462	1.0%	12,853	1.4%	O ₃
Dry cleaning (72)	101	.0%	179	.0%	5,319	.3%	132	.0%	28	.0%	*
Industrial sector total	3,560,059		1,813,927		1,261,612		2,454,578		909,383		

SOURCE.—EPA, Sector Notebook Project (1995).
NOTE.—For each pollutant, emissions in col. 1 and the number of short tons emitted per year. Share in col. 2 reports the fraction of industrial sector emissions. The paper's analysis designates an industry an emitter of a pollutant if it accounts for at least 7 percent of industrial sector emissions. Each industry's emission status is summarized in col. 3. Nitrogen dioxide and volatile organic compounds are the primary pollutants of concern (VOCs). If an industry emitted more than 7 percent of either of these pollutants, it is designated as O₃ emitter. The

(a) Table A1: National Ambient Air Quality Standards. (b) Table A2: Industrial pollutant releases by industry.

8 Short summary

- The paper studies the production costs of the Clean Air Act Amendments.
- Treatment is pollutant-specific nonattainment status interacted with industry pollutant emissions.
- The data are plant-level Census of Manufactures observations from 1967 to 1987.
- Nonattainment regulation reduced employment growth, capital growth, and shipment growth.
- The preferred estimates imply large aggregate manufacturing losses from 1972 to 1987.
- The design relies on rich fixed effects and pollutant-specific variation, but nonrandom regulatory assignment remains the central identification challenge.

9 Conclusion

9.1 Core conclusion

Greenstone finds that Clean Air Act regulation had meaningful economic costs for regulated manufacturing plants. Plants in nonattainment counties that emitted the relevant pollutant experienced slower growth in employment, capital stock, and shipments.

9.2 Economic intuition

Nonattainment designation imposes stricter permitting and emissions controls. These rules raise the cost of operating and expanding emitting plants. Firms respond by reducing employment, investment, and output at regulated locations.

9.3 Contribution revisited

The paper's incremental contribution is to quantify environmental regulation costs using plant-level data and pollutant-specific regulatory variation. This makes the analysis more credible and more policy-relevant than aggregate industry comparisons.

9.4 Policy interpretation

The paper does not measure the benefits of cleaner air. It estimates production-side costs. A full welfare analysis would compare these costs with health, mortality, amenity, and environmental benefits.

9.5 Limitations

Nonattainment status is not randomly assigned, and regulated counties differ from unregulated counties. The paper addresses this with fixed effects and robustness checks, but unobserved local shocks could still matter.

9.6 Takeaway

Environmental regulation can generate large local production costs. The Clean Air Act improved air quality, but it also reduced manufacturing activity in regulated areas. Understanding regulation requires measuring both sides of the cost-benefit ledger.